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Protocol generation date:
No.:

LABORATORY TEST PROTOCOL

Patient code:
Age:
Sex:

Sample type: dried blood spot (DBS)
Sample collected:
Sample received:
Analysis date:

Test	Result	Interpretation
Morning cortisol level	nmol/L	
Evening cortisol level	nmol/L	
Morning-to-evening cortisol ratio		

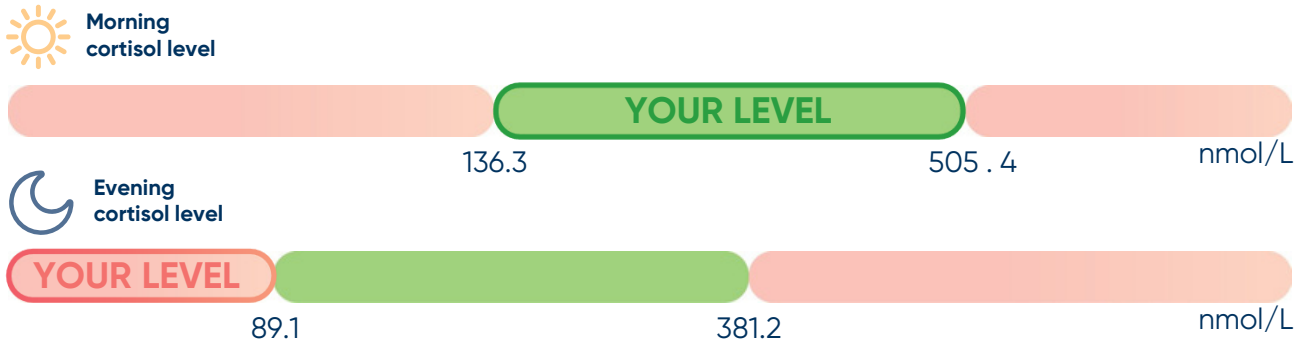
Analysis method: immunoassay
Employee license No.

Understanding your cortisol test results



Thank you for completing your cortisol test!

Results interpretation



Reference ranges were derived from manufacturer-provided validation data and were calculated from the samples collected in the morning (AM) and evening (PM) across racially diverse population.

Your cortisol follows a healthy daily rhythm, with higher levels in the morning and lower levels in the evening. This pattern indicates effective activation during the day and appropriate downregulation in the evening.

However, your evening cortisol level is below the reference range, suggesting a stronger-than-average suppression of cortisol at night. This finding is generally considered normal and reflects a healthy transition from daytime activity to evening recovery. It indicates that your body is able to let go of daytime stress and naturally shift into rest in the evening. This pattern may be associated with easier relaxation and more restorative sleep, although individual experiences can vary [1].

Cortisol results should always be interpreted alongside your symptoms and overall clinical picture. If you continue to experience persistent fatigue, unrefreshing sleep, difficulty coping with stress, or other ongoing symptoms, a qualified healthcare provider can help determine whether additional evaluation is appropriate.

Understanding your AM:PM cortisol ratio



What does your morning-to-evening cortisol ratio mean?



Research suggests that a more clearly defined difference between your morning and evening cortisol levels is usually reflected in an AM/PM ratio above 2:1 [2]. Although this ratio is not used as a medical diagnostic marker, it can serve as a helpful guide. Ratios above 2:1 often point to a well-structured daily rhythm, while lower ratios usually indicate a flatter pattern, where morning and evening levels are almost equal.

In simple terms, an AM/PM ratio above 2:1 tends to reflect a healthy daily cortisol rhythm. A lower ratio may mean that the contrast between daytime activation and evening wind-down is less distinct, something that can be linked to feeling less energized in the morning or more alert later in the day. If your ratio is on the lower side, it's a good idea to pay a bit more attention to your daily cortisol-supportive habits and to keep an eye on how your rhythm changes over time.

If this pattern persists or is combined with symptoms that affect your well-being, such as ongoing fatigue, sleep disruption, or difficulty managing stress, it may be worth discussing your results with a healthcare professional for a more detailed evaluation.

Result-based recommendations

If you notice early evening fatigue, reduced evening energy, or changes in sleep quality, it may be helpful to further support circadian alignment by maintaining a consistent sleep-wake schedule, spending time in natural morning light, and minimizing exposure to bright or blue light in the evening. Gentle evening routines that promote relaxation, such as calm movement, breathing exercises, or meditation, can further support healthy cortisol regulation. Gut health also plays a meaningful role in cortisol balance. Prebiotic foods nourish beneficial gut bacteria and may help lower cortisol levels, while probiotic-rich foods such as kefir, kombucha, sauerkraut, kimchi, and live yogurt introduce helpful microbes that support digestion, immunity, and overall wellbeing [3]. Together, these gentle lifestyle shifts create a strong foundation for a healthier cortisol rhythm and steadier energy throughout the day.

If you feel well, sleep soundly, and maintain stable daytime energy, this cortisol pattern is generally considered consistent with healthy recovery. Continue prioritizing consistent sleep, morning light exposure, balanced nutrition, and stress-reducing practices.



Here's a quick explanation of what cortisol is and why it matters.

Cortisol is often misunderstood as simply a “stress hormone” that tells us if we are stressed. Don’t be mistaken, it’s not as simple as it seems. First, a single measure of cortisol will tell us nothing since cortisol levels go up and down within minutes [4]. Second, cortisol is necessary for the correct functioning of almost every part of the body [5]. A normal cortisol response is not harmful, on the contrary, it helps us adapt to challenges, coordinate our metabolism, and stay in sync with our daily activity and sleep patterns [4]. Balanced cortisol levels are important for maintaining overall well-being and supporting the body’s ability to adapt to physical and psychological challenges.

Cortisol plays an important role in learning, brain plasticity, and immune function. Its natural daily rise supports the formation and clearing of neural connections, helping the brain learn and adapt. The temporary rise of cortisol during stress, helps activate the adaptive immune system so the body can respond to threats such as infections or injuries. For example, an appropriate cortisol response affects how effectively the body reacts to a vaccination or heals a wound [4,6].

So cortisol itself is not the “bad guy.” It becomes problematic only when its levels stay too high when they should be low, or remain too low when the body needs them. Cortisol works best when its natural circadian rhythm is intact.

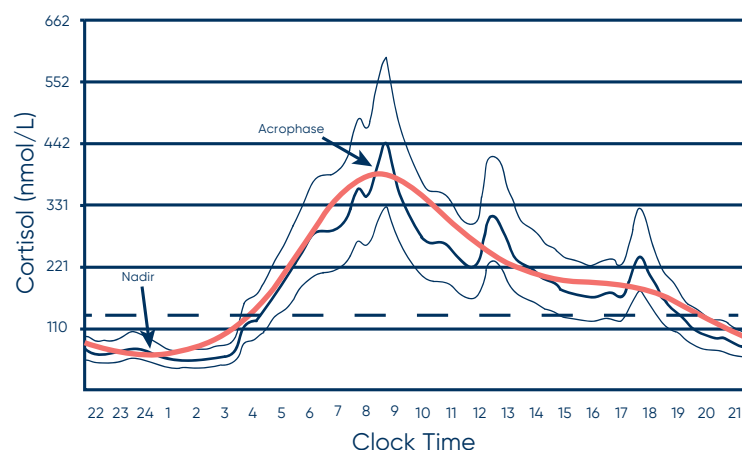


Figure 1. The physiological circadian pattern of cortisol release, adapted according to [7]

Cortisol follows a predictable daily rhythm: it rises sharply in the morning, supports energy during the day, and gradually falls toward the evening.

This strong morning rise is guided by the body’s internal circadian clock in the brain, known as the suprachiasmatic nucleus (SCN)¹, which coordinates daily hormone rhythms, including cortisol release. Exposure to natural morning light reinforces this process, helping the body transition from sleep to wakefulness [8]. Cortisol levels fall to their lowest point in the evening and during the early phases of sleep, supporting the body’s shift into rest and recovery. The process of falling asleep actively suppresses cortisol secretion, maintaining the low levels essential for nighttime physiology [9].

¹ The suprachiasmatic nucleus (SCN), located in the hypothalamus, receives light input from the retina and coordinates daily biological rhythms by regulating the activity of the hypothalamic-pituitary-adrenal (HPA) axis (doi: 10.3389/fnbeh.2020.601939)

As the final step, take a moment to relax and allow your body to settle into its natural rhythm.



References:

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